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Course: CSA0421 - Operating System | **GitHub Repo:** krishshiva721/CSA04--Assignment1

TASK 12: DAY-4 DESIGN CHALLENGE ADAPTATION

Remote IoT Home Monitoring & Hybrid Cloud Database Re-Evaluation

A. IMPACT ANALYSIS & SYSTEM SCOPE (2 MARKS)

The introduction of remote home monitoring devices and a cloud-based record repository expands the security perimeter beyond the hospital LAN. Operating system services impacted include network communications (enforcing mandatory TLS 1.3 / WireGuard tunneling), hardware-rooted cryptographic authentication (TPM 2.0 / Secure Boot), and fault-tolerant network queuing services.

B. EVALUATION OF ORIGINAL DESIGN LIMITATIONS (2 MARKS)

The original local Active-Passive database cluster cannot scale to federate record sharing across partner hospitals without introducing WAN latency bottlenecks. Furthermore, home monitoring devices operating over public cellular/broadband networks face unbounded Internet jitter, preventing strict 10 ms hard real-time latency across the WAN.

C. ADAPTATION ARCHITECTURE: EDGE-CLOUD MODEL (2 MARKS)

System Component	Revised Architecture	OS Security & Network Adaptation
Remote Home Monitors	Embedded RTOS (FreeRTOS / Zephyr)	Local alarm evaluation (< 10 ms alarm speaker). Asynchronous cloud telemetry push over MQTTS with local buffering.
Shared EHR Repository	Distributed Cloud OS (Kubernetes / CockroachDB)	Zero-Trust architecture with mutual TLS (mTLS), OAuth2 JWT authorization, and encrypted storage volume drivers.

D. EVIDENCE-BASED DEFENSE & TRADE-OFF EVALUATION (2 MARKS)

> **Architectural Defense: Dual-Loop Processing Strategy**
>
> We adopt a **MODIFIED Edge-Cloud Design**. The safety deadline for patient alarms is maintained by performing local evaluation on the remote device's embedded RTOS, raising audible/visual alarms in the patient's home within `< 5 ms` (Loop 1). Emergency telemetry is simultaneously queued over cellular networks to the Cloud central dashboard (Loop 2). This eliminates reliance on WAN network availability for life safety while leveraging cloud elasticity for shared records.